

PEAWANUCK, ON, Canada

WMO: 714340

Lat: 54.990N

Lon: 85.441W

Elev: 173

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.2	-30.9	-42.6	0.5	-34.8	-38.3	0.6	-30.8	20.7	2.5	18.9	1.7	5.5	250	0.810

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	81.8	64.1	77.7	62.3	73.4	60.8	67.2	76.4	64.7	73.4	62.3	70.6	12.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	89.5	71.3	61.3	81.7	68.3	58.7	74.1	66.6	31.9	76.6	29.9	72.8	28.2	70.7	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	18.3	16.4	DB	-41.5	89.6	4.8	4.6	-44.9	92.9	-47.8	95.6	-50.5	98.2	-54.0	101.6	(4)
(5)			WB	-41.5	72.1	4.6	2.8	-44.8	74.2	-47.6	75.8	-50.2	77.4	-53.5	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.1	-9.7	-7.3	5.0	21.4	37.0	49.8	56.9	56.1	47.5	35.7	18.5	0.1		
	DBStd	26.06	13.00	11.86	14.15	12.14	10.68	9.93	8.44	7.69	7.80	7.34	11.65	15.21		
	HDD50	9393	1852	1604	1395	858	434	125	23	22	139	448	945	1548		
	HDD65	14264	2317	2024	1860	1307	870	469	280	295	526	908	1395	2013		
	CDD50	669	0	0	0	0	30	119	238	212	65	5	0	0		
	CDD65	65	0	0	0	0	1	13	29	20	2	0	0	0		
	CDH74	808	0	0	0	0	32	205	344	209	18	0	0	0		
	CDH80	199	0	0	0	0	3	52	85	57	2	0	0	0		
	WSAvg	8.5	7.9	8.0	8.5	8.9	9.0	8.6	8.2	8.4	8.8	8.8	8.6	8.1		
	PrecAvg	22.8	1.1	0.8	1.1	1.2	2.1	2.5	3.0	2.8	2.8	2.2	1.9	1.2		
(15) Precipitation	PrecMax	28.4	1.9	1.8	2.0	2.2	3.5	5.9	5.1	4.1	6.1	3.6	3.3	3.2		
	PrecMin	18.6	0.4	0.3	0.3	0.4	1.0	0.7	1.4	1.3	1.1	0.7	0.9	0.5		
	PrecStd	2.6	0.4	0.4	0.4	0.5	0.7	1.1	0.9	0.7	1.1	0.7	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	32.2	44.5	57.6	78.9	85.8	87.4	86.4	76.2	63.5	45.5	32.4		
		MCWB	29.1	31.2	37.9	47.2	59.3	65.7	68.1	66.9	62.5	57.1	42.0	31.0		
	2%	DB	22.9	23.8	38.7	49.1	70.7	80.0	81.9	79.9	70.6	55.5	38.8	29.2		
		MCWB	22.3	22.2	34.9	40.6	55.2	61.9	65.1	64.7	60.4	50.0	36.5	28.5		
	5%	DB	16.2	17.9	33.4	43.6	62.9	74.5	77.8	74.5	65.4	50.3	35.2	26.2		
		MCWB	15.4	16.6	30.7	36.9	52.2	59.2	62.8	62.6	58.0	46.7	33.9	25.5		
	10%	DB	9.7	11.7	27.3	39.3	55.4	69.1	73.1	70.3	60.6	45.8	32.7	22.6		
		MCWB	8.9	10.3	24.7	34.2	47.6	57.6	61.3	60.7	54.5	42.9	31.8	21.8		
	0.4%	WB	29.2	31.5	38.7	47.6	60.6	70.3	71.8	69.4	65.5	57.3	42.8	31.4		
		MCDB	29.9	32.5	43.1	57.4	73.0	81.4	82.6	80.5	73.0	62.9	44.5	32.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	22.6	22.3	35.1	41.1	56.6	64.5	67.3	66.8	62.0	51.0	36.9	28.6		
		MCDB	23.2	23.6	38.4	48.2	69.0	73.0	76.3	76.5	68.3	54.5	38.6	29.4		
	5%	WB	15.1	16.9	30.8	37.5	52.5	60.9	64.6	64.2	58.7	46.9	34.1	25.2		
		MCDB	15.8	18.1	33.5	43.0	62.5	72.3	73.8	72.4	64.8	50.1	35.2	25.7		
	10%	WB	8.5	10.6	24.8	34.3	47.8	57.8	62.3	61.5	55.2	43.3	32.0	21.3		
		MCDB	9.2	11.8	27.2	38.9	56.1	69.4	71.0	68.6	59.8	46.0	32.8	22.1		
	5% DB	MDBR	16.7	20.4	22.3	19.9	18.3	23.1	22.2	19.1	15.4	10.5	11.8	14.4		
		MCDBR	18.8	22.1	24.0	23.5	30.0	30.7	27.8	25.9	23.3	17.5	10.8	12.8		
		MCWBR	19.1	20.7	20.7	16.5	16.9	15.7	14.1	13.8	14.4	13.0	9.7	12.2		
		MCWBR	19.2	22.0	22.7	21.9	29.1	27.8	24.0	22.5	20.7	16.6	10.9	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.253	0.288	0.369	0.378	0.374	0.395	0.372	0.343	0.303	0.273	0.221		
		taud	2.191	2.222	2.178	2.038	2.239	2.353	2.356	2.423	2.499	2.530	2.244	2.165		
	Ebn at Noon		229	267	280	268	274	277	267	267	262	249	207	206		
		Edh at Noon	18	26	35	46	40	36	36	32	26	20	17	15		
(44) All-Sky Solar Radiation	RadAvg		290	638	1129	1634	1726	1818	1687	1296	855	462	217	183		
	RadStd		26	44	78	109	166	208	147	112	89	41	33	28		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-74	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-74	N/A

Nomenclature: See separate page